Exact Tangent Moment Operators for a Markov-Switching Hénon Cocycle

Route-A finite certificate C117

Abstract

We freeze a two-state Markov cocycle of dissipative polynomial Hénon maps sharing the origin. Once the environment chronology is fixed, the tangent cocycle owns exact operators on environment-conditioned first and symmetric second moments, of dimensions four and six. We give their matrices, traces through power six, and full determinant polynomials over $\mathbb{Q}$. A stationary control shows that averaging and taking a symmetric square differ by a rank-one matrix. These are local finite moment owners, not a global nonlinear Fredholm construction.

1 Frozen random dynamics and chronology

Consider

$$F_0(x,y) = (x^2 + \tfrac{1}{2}x - \tfrac{1}{3}y, x), \qquad F_1(x,y) = (x^2 - x - \tfrac{1}{2}y, x),$$

with transition matrix

$$P = \begin{pmatrix} 2/3 & 1/3 \\ 1/4 & 3/4 \end{pmatrix}, \qquad \pi = (3/7, 4/7).$$

Rows of P are the old mode and columns the new mode. Our load-bearing convention is: $s_n = i$ changes to $s_{n+1} = j$ with probability P_{ij}, then F_j is applied. Both maps fix the origin, and their tangent matrices are

$$B_0 = \begin{pmatrix} 1/2 & -1/3 \\ 1 & 0 \end{pmatrix}, \qquad B_1 = \begin{pmatrix} -1 & -1/2 \\ 1 & 0 \end{pmatrix}, \qquad (\det B_0, \det B_1) = (1/3, 1/2).$$

Every statement below concerns this common-fixed-point tangent cocycle.

2 Source-owned conditional moment operators

Let $m_i = \mathbb{E}[(x,y)^\mathsf{T} \mathbf{1}_{s=i}]$. The exact update is

$$m'_j = \sum_i P_{ij} B_j m_i,$$

so the block $[j,i]$ of the first-moment owner is $P_{ij}B_j$. In the basis $(0{:}x, 0{:}y, 1{:}x, 1{:}y)$ this is

$$A_1 = \begin{pmatrix} 1/3 & -2/9 & 1/8 & -1/12 \\ 2/3 & 0 & 1/4 & 0 \\ -1/3 & -1/6 & -3/4 & -3/8 \\ 1/3 & 0 & 3/4 & 0 \end{pmatrix}.$$

For $(x', y') = (ax - by, x)$, the pullback on (x^2, xy, y^2) is

$$S(a,b) = \begin{pmatrix} a^2 & -2ab & b^2 \\ a & -b & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

Replacing B_j by $S_j = S(a_j, b_j)$ gives the six-dimensional operator $[A_2]_{j,i} = P_{ij} S_j$. Thus the spaces and actions precede all coefficient calculations; neither matrix is fitted to a target sequence.

3 Exact trace and determinant data

The first six traces are

$$\begin{aligned}\mathrm{tr}(A_1^n):&\ -5/12, -11/27, 95/384, -35249/373248,\\ &\ -130525/8957952, 3363181/80621568,\\ \mathrm{tr}(A_2^n):&\ 23/72, 269/1728, 110897/373248, 1465921/8957952,\\ &\ 152582093/1934917632, 2458516129/46438023168.\end{aligned}$$

Independent determinant expansion and Newton reconstruction agree exactly:

$$\begin{aligned}\det(I-zA_1)&=1+\tfrac{5}{12}z+\tfrac{251}{864}z^2+\tfrac{25}{1728}z^3+\tfrac{25}{864}z^4,\\ \det(I-zA_2)&=1-\tfrac{23}{72}z-\tfrac{139}{5184}z^2-\tfrac{29713}{373248}z^3\\ &\qquad-\tfrac{14605}{1492992}z^4-\tfrac{925}{1492992}z^5+\tfrac{125}{373248}z^6.\end{aligned}$$

4 A stationary averaging control

Stationary averaging gives

$$\bar{B}=\begin{pmatrix}-5/14 & -3/7\\ 1 & 0\end{pmatrix}.$$

However, with $\bar{S}=\sum_j \pi_j S_j$,

$$\bar{S}-\mathrm{Sym}^2(\bar{B})=\begin{pmatrix}27/49 & 6/49 & 1/147\\ 0 & 0 & 0\\ 0 & 0 & 0\end{pmatrix},$$

a nonzero rank-one gap. The exact quadratic moment lift therefore retains a switching-variance term discarded by the naive averaged-map control.

5 Reproducibility and boundary

From the package directory run

```
python3 code/c117_markov_producer.py
python3 code/c117_markov_checker.py
python3 code/c117_sympy_crosscheck.py
python3 code/c117_replay.py
python3 code/c117_mutation.py
```

The independent checker reconstructs both operators; SymPy performs 26 exact Newton/trace checks, canonical replay passes, and all 12 hostile semantic mutations are rejected.

The honest verdict is $A1 =$ `A1_WEAK` and $A2 =$ `A2_CERTIFIED_PREFIX`, qualified as source-owned finite tangent-moment operators only; $A3$ is not addressed and $A4$ fails. We do not claim a complete nonlinear random-orbit atlas, a global nonlinear transfer/Fredholm/nuclear owner, arithmetic local data, Euler factors, root numbers, automorphy, a Hilbert–Pólya operator, or Route B. The literal firewall is `NO_BAD_EULER_OR_ROOT_NUMBER`.